

BÙI HỒNG PHÚC (Ethan)

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PROFESSIONAL SUMMARY

Data professional with strong SQL and Python skills and a proven record of turning raw, unstructured data — including financial and identity documents — into measurable business outcomes. Comfortable across the full analytic workflow: extracting and cleaning data, feature engineering, building and validating models, and presenting results to non-technical stakeholders. Currently self-studying credit-risk fundamentals, aiming to build a long-term career in data science within consumer finance.

Experience

NData Science Research Assistant — NLP Document AI Lab, University of Science *Feb 2025 – Present*

- Built an automated data-extraction pipeline (OCR + computer vision) that converts financial documents — forms, invoices — into structured, query-ready data, cutting manual processing time by 60% and improving extraction accuracy by 20%.
- Created data-validation and annotation tooling to ensure clean, representative training data, improving team throughput by 30% and reducing downstream model errors.
- Applied and fine-tuned machine-learning models on Vietnamese financial text for translation, summarization, and information retrieval, strengthening the lab's internal knowledge-retrieval system.
- Mentored 3 junior members, owned the team's Git codebase, and introduced a code-review workflow that reduced integration bugs by 15%.

Technology Development Intern — CMT Global *Feb 2025 – Apr 2025*

- Designed and ran large-scale data-collection (web-scraping) pipelines that aggregated market data and structured raw JSON/HTML into a query-ready SQL database for business intelligence and reporting.
- Resolved critical production issues to maintain 99% system stability for the analytics team.
- Performed functional and manual testing for product verification, reducing post-release defects by 25%.

SELECTED PROJECTS

Enterprise Database & Reporting Automation (AI Agent) *May 2026*

- Built an AI agent that automates SQL querying and financial-report generation across enterprise databases, cutting manual data-retrieval time by 40% and accelerating data-driven decisions.
- Integrated the agent with internal dashboards and led a 2-person team to define business use cases with cross-functional stakeholders.
- Tech: OpenClaw, Python, SQL, LangChain, LLM APIs

KYC Document Understanding for Fintech — CoverGO Hackathon 2025 *Oct 2025 – Nov 2025*

- Built a system to automatically extract key fields from Vietnamese financial and identity documents (invoices, ID cards), simulating a practical KYC solution for fintech.
- Tuned document-understanding models to parse complex table structures, reaching 92% F1 on a held-out test set; designed semi-automated annotation to speed up iteration.
- Tech: Python, PyTorch, OCR/VLM, Gradio
- **Project:** github.com/Phucgiacat/CoverHackatho-project 

Model Reliability for Risk-Sensitive Applications — Data Science Challenge *Aug 2025 – Oct 2025*

- Designed a reward model to detect factual errors in model outputs, targeting reliability for risk-sensitive domains such as finance.
- • Built automated evaluation pipelines to measure error rates across model versions, reaching 86% accuracy

and reducing false positives by 18%.

- **Project:** github.com/monster1909/Detect-hallucination-using-LoRa-and-GRPO [↗](#)

End-to-End Data Labeling Pipeline (Published) — Personal Project

Feb 2024 – Aug 2024

- • Built an automated data-labeling system (OCR + computer vision) reaching 90% character-level accuracy and cutting annotation workload by 60%.
- • Implemented an error-correction algorithm (Levenshtein matching) to improve data quality for downstream models;
- Published a paper on the methodology at **VCL 2025** (Vietnamese Conference on Linguistics).
- **Project:** github.com/Phucgiacat/ocr_corrector [↗](#)

Education

University of Science, Ho Chi Minh City

Sep 2022 – Present
(Expected 2026)

- Bachelor of Information Technology
- **GPA: 3.2/4.0** (Major GPA: 3.6/4.0 in AI/ML courses)
- **Relevant Coursework:** Probability Statistics, Machine Learning, Data Mining, Database Systems, Deep Learning

Certifications & Achievements

- 2nd Prize – Bosch Innovation Day 2025
- Coursera: Supervised ML & Advanced Learning Algorithms – DeepLearning.AI (2024)
- Python (Basic) – HackerRank (2024)
- 3rd Prize – Provincial Science Fair
- Participant – UIT Data Science Challenge 2025
- **Publication:** Paper accepted at VCL 2025 (Vietnamese Conference on Linguistics)
- [Online Portfolio & Certificates Available](#) [↗](#)

Technical Skills

Data & Analytics: SQL, Python (pandas, NumPy), exploratory data analysis, feature engineering, statistical modeling, reporting & dashboards

Machine Learning: scikit-learn, PyTorch; model training, validation & performance evaluation. LightGBM, XGBoost

Visualization & BI: Matplotlib, dashboard building & business reporting (Power BI: in progress)

Tools & Workflow: Git, Docker, REST APIs; cross-functional collaboration & stakeholder communication

Domain: Credit-risk fundamentals (self-taught), KYC / financial-document processing